Number & Algebra

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the quadratic function $f(x) = x^2 + bx + c$. The graph of $f(x)$ is shown below. The vertex of the parabola is at $(2, -5)$ and it passes through the point $(0, -1)$. Determine the values of b and c.

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(2, -5)$, we have $f(x) = a(x - 2)^2 - 5$. Since the parabola passes through $(0, -1)$, we can substitute these values into the equation: $-1 = a(0 - 2)^2 - 5$ $-1 = a(4) - 5$ $4 = 4a$ $a = 1$. So, the function in vertex form is $f(x) = 1(x - 2)^2 - 5$. Expanding this, we get: $f(x) = (x^2 - 4x + 4) - 5$ $f(x) = x^2 - 4x - 1$. Comparing this to the given form $f(x) = x^2 + bx + c$, we can identify the coef.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The graph of the quadratic function $f(x) = 2x^2 - kx + 5$ passes through the point (3, 14). Find the value of k.

The graph of the function $f(x) = 2x^2 - kx + 5$ passes through the point (3, 14). This means that when $x = 3$, $f(x) = 14$. We can substitute these values into the function's equation: $14 = 2(3)^2 - k(3) + 5$ $14 = 2(9) - 3k + 5$ $14 = 18 - 3k + 5$ $14 = 23 - 3k$ Now, we need to solve for k: $3k = 23 - 14$ $3k = 9$ $k = \frac{9}{3}$ $k = 3$

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The diagram shows the graph of the function $f(x) = ax^2 + bx + c$. The graph passes through the points $(1, -4)$, $(2, -5)$, and $(4, -1)$. Determine the value of a.

Since the graph of $f(x) = ax^2 + bx + c$ passes through the given points, we can form a system of linear equations by substituting the coordinates of these points into the function's equation: 1. For $(1, -4)$: $a(1)^2 + b(1) + c = -4 \Rightarrow a + b + c = -4$ (Equation 1) 2. For $(2, -5)$: $a(2)^2 + b(2) + c = -5 \Rightarrow 4a + 2b + c = -5$ (Equation 2) 3. For $(4, -1)$: $a(4)^2 + b(4) + c = -1 \Rightarrow 16a + 4b + c = -1$ (Equation 3) Now, we solve this system of equations. Subtract Equation 1 from Equatio.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the two functions $f(x) = \log_2(x)$ and $g(x) = 3 - x$. The graphs of $f(x)$ and $g(x)$ intersect at a point P, and $f(x)$ intersects the x-axis at point Q. A line segment is drawn from Q to a point R on the graph of $g(x)$ such that QR is perpendicular to the x-axis. Find the coordinates of the point R.

The graph of $f(x) = \log_2(x)$ intersects the x-axis when $f(x) = 0$. Thus, $\log_2(x) = 0$, which implies $x = 2^0 = 1$. So, point Q has coordinates (1, 0). The line segment QR is perpendicular to the x-axis, meaning that the x-coordinate of R is the same as the x-coordinate of Q. Therefore, the x-coordinate of R is 1. Point R lies on the graph of $g(x) = 3 - x$. Substituting the x-coordinate of R into the equation for $g(x)$, we get $g(1) = 3 - 1 = 2$. Thus, the coordinates of point R are (1, 2). Let's check the.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = -x^2 + 6x - 5$. The graph of this function is shown below. A tangent line to the graph at point $P(2, 3)$ intersects the x-axis at point Q. Determine the coordinates of point Q.

To find the equation of the tangent line, we first need the derivative of the function $f(x) = -x^2 + 6x - 5$. The derivative is $f'(x) = -2x + 6$. The slope of the tangent line at point $P(2, 3)$ is $f'(2) = -2(2) + 6 = -4 + 6 = 2$. Using the point-slope form of a line, $y - y_1 = m(x - x_1)$, with $(x_1, y_1) = (2, 3)$ and $m = 2$, the equation of the tangent line is $y - 3 = 2(x - 2)$. This simplifies to $y - 3 = 2x - 4$, or $y = 2x - 1$. To find the x-intercept (point Q), we set $y = 0$: $0 = 2x - 1$. Solving.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = x^2 - 4x + 3$ and the line $g(x) = x - 1$. The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.

To find the intersection points, we set the two functions equal to each other: $f(x) = g(x)$ $x^2 - 4x + 3 = x - 1$ $x^2 - 5x + 4 = 0$ Factoring the quadratic equation: $(x - 1)(x - 4) = 0$ This gives us two possible x-values for the intersection points: $x = 1$ or $x = 4$. Using $g(x) = x - 1$ to find the corresponding y-values: For $x = 1$: $y = 1 - 1 = 0$. The intersection point is $(1, 0)$. For $x = 4$: $y = 4 - 1 = 3$. The intersection point is $(4, 3)$. Looking at the graph, one of the intersection points is $(1, 0)$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = x^2 - 2x + 1$. Determine the y-coordinate of the vertex of the graph of $y = f(x)$.

The graph of a quadratic function $f(x) = ax^2 + bx + c$ is a parabola. The x-coordinate of the vertex is given by $x = -\frac{b}{2a}$. For the function $f(x) = x^2 - 2x + 1$, we have $a = 1$ and $b = -2$. Therefore, the x-coordinate of the vertex is $x = -\frac{-2}{2(1)} = \frac{2}{2} = 1$. To find the y-coordinate of the vertex, we substitute this value of x back into the function: $f(1) = (1)^2 - 2(1) + 1 = 1 - 2 + 1 = 0$. Thus, the y-coordinate of the vertex is 0.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = -x^2 + 6x - 5$. Determine the coordinates of the vertex of the graph of $y = f(x)$.

The given function is a quadratic of the form $f(x) = ax^2 + bx + c$, where $a = -1$, $b = 6$, and $c = -5$. The x-coordinate of the vertex of a parabola is given by $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{6}{2(-1)} = -\frac{6}{-2} = 3$. To find the y-coordinate of the vertex, substitute $x = 3$ into the function: $f(3) = -(3)^2 + 6(3) - 5 = -9 + 18 - 5 = 9 - 5 = 4$. Therefore, the coordinates of the vertex are (3, 4).

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph passes through the point $(5, -2)$. Find the value of k.

The function is given by $f(x) = x^2 - 6x + k$. We are given that the graph of this function passes through the point $(5, -2)$. This means that when $x = 5$, $f(x) = -2$. Substituting these values into the function, we get: $-2 = (5)^2 - 6(5) + k$ $-2 = 25 - 30 + k$ $-2 = -5 + k$ To solve for k, we add 5 to both sides of the equation: $-2 + 5 = k$ $3 = k$ Therefore, the value of k is 3.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = e^{-x^2}$ and the line $y = k$. The graph of $y = f(x)$ is concave up for $x \in (-\infty, -a) \cup (a, \infty)$ and concave down for $x \in (-a, a)$. Find the exact value of a^2.

To determine the intervals of concavity, we need to find the second derivative of $f(x) = e^{-x^2}$. First derivative: $f'(x) = e^{-x^2} \cdot (-2x) = -2xe^{-x^2}$. Second derivative using the product rule: $f''(x) = (-2)e^{-x^2} + (-2x) \cdot e^{-x^2} \cdot (-2x)$ $f''(x) = -2e^{-x^2} + 4x^2e^{-x^2} = e^{-x^2}(4x^2 - 2)$. The concavity of $f(x)$ is determined by the sign of $f''(x)$. Since e^{-x^2} is always positive, the sign of $f''(x)$ is determined by the sign of $4x^2 - 2$. The function is concave down when.

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11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The depth of water $h(t)$ in meters at a harbor entrance t hours after midnight is modelled by $h(t) = 3.5\cos\left(\frac{\pi t}{6}\right) + 6.5$ for $0 \le t \le 24$. (a) State the maximum and minimum depths of water and their times of occurrence. (b) Find the period of the tidal oscillation. (c) A cargo ship requires a minimum water depth of 8.25 meters to safely enter the harbor. Find the total number of hours during the first 12 hours of the day when the ship can safely enter.

(a) Maximum depth occurs when $\cos = 1$: $h_{\max} = 6.5 + 3.5 = 10.0$ m. This occurs at $t = 0$, $t = 12$, and $t = 24$ hours. Minimum depth occurs when $\cos = -1$: $h_{\min} = 6.5 - 3.5 = 3.0$ m. This occurs at $t = 6$ and $t = 18$ hours. (b) Period $T = \frac{2\pi}{B} = \frac{2\pi}{\pi/6} = 12$ hours. (c) Solve $h(t) \ge 8.25$ over $[0, 12]$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = x^3 - 6x^2 + 5$. Find the exact value of the sum of the x-coordinates of the local maximum and local minimum points of the graph of $y = f(x)$.

To find the local maximum and minimum points, we first need to find the critical points by setting the first derivative of $f(x)$ to zero. First derivative: $f'(x) = \frac{d}{dx}(x^3 - 6x^2 + 5) = 3x^2 - 12x$. Set $f'(x) = 0$ to find critical points: $3x^2 - 12x = 0$ $3x(x - 4) = 0$ So, the critical points are $x = 0$ and $x = 4$. To determine whether these are local maxima or minima, we can use the second derivative test. Second derivative: $f''(x) = \frac{d}{dx}(3x^2 - 12x) = 6x - 12$. Evaluate $f''(x)$ at the cr.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The function $f(x) = \frac{2x-5}{x+3}$ is defined for $xeq -3$. (a) Find the coordinates of the axes intercepts of the graph of $y = f(x)$. (b) Write down the equations of the vertical and horizontal asymptotes. (c) Find an expression for $f^{-1}(x)$ and state its domain and range.

(a) y-intercept (at $x = 0$): $f(0) = \frac{-5}{3} = -\frac{5}{3}$, giving $\left(0, -\frac{5}{3}\right)$. x-intercept (at $y = 0$): $2x - 5 = 0 \Rightarrow x = 2.5$, giving $(2.5, 0)$. (b) Vertical asymptote where denominator is zero: $x = -3$. Horizontal asymptote: $y = \lim_{x \to \pm\infty} \frac{2x-5}{x+3} = 2$. (c) Set $y = \frac{2x-5}{x+3} \Rightarrow y(x + 3) = 2x - 5 \Rightarrow xy + 3y = 2x - 5$. Rearrange.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the triangle ABC shown, point D lies on AB and point E lies on AC such that DE is parallel to BC. Given that AD = 6, DB = 4, and the area of triangle ADE is 27 square units, find the area of triangle ABC.

Since DE is parallel to BC, by the property of similar triangles, triangle ADE is similar to triangle ABC. The ratio of corresponding sides is $\frac{AD}{AB} = \frac{AD}{AD+DB} = \frac{6}{6+4} = \frac{6}{10} = \frac{3}{5}$. For similar triangles, the ratio of their areas is the square of the ratio of their corresponding sides. Therefore, $\frac{\text{Area}(ADE)}{\text{Area}(ABC)} = \left(\frac{AD}{AB}\right)^2 = \left(\frac{3}{5}\right)^2 = \frac{9}{25}$. We are given that the area of triangle ADE is 27 square.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 AI SL — Specific Topics

The box plot below shows the distribution of scores for a class on a recent statistics quiz. The five-number summary is given by minimum, first quartile (Q1), median, third quartile (Q3), and maximum. If the interquartile range (IQR) is 25, and the median score is 70, find the value of Q1.

The interquartile range (IQR) is defined as the difference between the third quartile (Q3) and the first quartile (Q1), i.e., $\text{IQR} = Q_3 - Q_1$. We are given that the IQR is 25. The median of the dataset is given as 70. In a box plot, the median is represented by the line inside the box. Therefore, $Q_2 = 70$. The box plot also indicates that the third quartile (Q3) is 70. This is because the right edge of the box aligns with the median line, indicating $Q_3 = 70$. Now we can use the IQR formula to find Q1.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Functions

Evaluate the definite integral $\int_0^{\sqrt{7}} \frac{4x}{\sqrt{x^2+9}}\,dx$ using the substitution $u = x^2 + 9$.

1. Let $u = x^2 + 9$. Then $du = 2x\,dx \Rightarrow 4x\,dx = 2\,du$. 2. Transform the limits of integration: When $x = 0$: $u = 0^2 + 9 = 9$. When $x = \sqrt{7}$: $u = (\sqrt{7})^2 + 9 = 7 + 9 = 16$. 3. Rewrite the integral in terms of u: $\int_9^{16} \frac{2}{\sqrt{u}}\,du = 2\int_9^{16} u^{-1/2}\,du = 2[2u^{1/2}]_9^{16} = 4[\sqrt{u}]_9^{16}$. 4. Evaluate at the boundaries: $4(\sqrt{16} - \sqrt{9}) = 4(4 - 3) = 4(1) = 4$. The exact value of the integral is 4.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Geometry & Trigonometry

Solve the trigonometric equation $2\sin^2\theta + 3\cos\theta - 3 = 0$ for $0 \le \theta \le 2\pi$, giving exact radian answers.

1. Substitute $\sin^2\theta = 1 - \cos^2\theta$: $2(1 - \cos^2\theta) + 3\cos\theta - 3 = 0 \Rightarrow 2 - 2\cos^2\theta + 3\cos\theta - 3 = 0$. $-2\cos^2\theta + 3\cos\theta - 1 = 0 \Rightarrow 2\cos^2\theta - 3\cos\theta + 1 = 0$. 2. Factor the quadratic in $\cos\theta$: $(2\cos\theta - 1)(\cos\theta - 1) = 0$. 3. Solve each branch: Branch 1: $2\cos\theta - 1 = 0 \Rightarrow \cos\theta = \frac{1}{2}$. Reference angle: $\alpha = \frac{\pi}{3}$. In $[0, 2\pi]$, cosine is positive in Q1 and Q4.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Statistics & Probability

Consider the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at points A and B. One of these intersection points is $(1, 0)$. Find the coordinates of the other intersection point.

To find the intersection points of the two graphs, we set $f(x) = g(x)$. $x^2 - 4x + 3 = x - 1$ Subtract x from both sides: $x^2 - 5x + 3 = -1$ Add 1 to both sides: $x^2 - 5x + 4 = 0$ Factor the quadratic equation: $(x - 1)(x - 4) = 0$ This gives us two possible x-coordinates for the intersection points: $x = 1$ and $x = 4$. We are given that one intersection point has an x-coordinate of 1. To find the corresponding y-coordinate, we can substitute $x = 1$ into either $f(x)$ or $g(x)$. Using $g(x) = x - 1$, we get.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Calculus

Solve the logarithmic equation $\log_3(x+5) + \log_3(x-1) = 2 + \log_3(2)$ for all real values of x.

1. Domain constraints: $x+5>0 \Rightarrow x>-5$ and $x-1>0 \Rightarrow x>1$. Thus $x>1$. 2. Rewrite the right-hand side using $2 = \log_3(3^2) = \log_3(9)$: $2 + \log_3(2) = \log_3(9) + \log_3(2) = \log_3(18)$. 3. Combine the left-hand side: $\log_3((x+5)(x-1)) = \log_3(18) \Rightarrow (x+5)(x-1) = 18$. 4. Expand and solve the quadratic equation: $x^2 + 4x - 5 = 18 \Rightarrow x^2 + 4x - 23 = 0$. 5. Quadratic formula.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 AI SL — Specific Topics

A discrete random variable X follows a binomial distribution $X \sim B(15, 0.35)$. (a) Find $P(X=5)$. (b) Find $P(X \geq 4)$. (c) Calculate the mean $E(X)$ and standard deviation σ of X.

(a) $P(X=5) = \binom{15}{5}(0.35)^5(0.65)^{10} = 3003 \times (0.005252) \times (0.013463) \approx 0.2123$. (b) $P(X \geq 4) = 1 - P(X \leq 3)$. Using GDC `binomcdf(15, 0.35, 3)`: $P(X \leq 3) \approx 0.1727$. $P(X \geq 4) = 1 - 0.1727 = 0.8273$ (82.7%). (c) Mean: $E(X) = np = 15 \times 0.35 = 5.25$. Variance: $\text{Var}(X) = np(1-p) = 15 \times 0.35 \times 0.65 = 3.4125$. Standard deviation: $\sigma = \sqrt{3.4125} \approx 1.847 \approx 1.85$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in solve by factorisation, evaluate $f(2)$.

a. 7 ✓

b. 9

c. 4

d. 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x-2)(x-4) = 0$ yields the roots $x = 2$ and $x = 4$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in solve quadratics by completing the square, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in solve using the quadratic formula, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solve by factorisation consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solve quadratics by completing the square consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for discriminant of a quadratic consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in discriminant of a quadratic, evaluate $f(4)$.

a. 26 ✓

b. 31

c. 20

d. 30

Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for solve quadratics using various methods consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 7$ in solve quadratics using various methods, evaluate $f(4)$.

a. 31 ✓

b. 37

c. 24

d. 35

Substitute $x = 4$: $f(4) = 6(4) + 7 = 31$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 8$ in graph basic quadratics, evaluate $f(3)$.

a. 14 ✓

b. 16

c. 6

d. 18

Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the quadratic function $f(x) = x^2 - 5x + k$ is shown below. The graph passes through the point $(3, -2)$. Find the value of k.

a. **2** ✓

b. 5

c. 7

d. 9

The function is given by $f(x) = x^2 - 5x + k$. $We are given that the graph passes through the point (3, -2)$. This means that when $x = 3$, $f(x) = -2$. $Substitute these values into the function equation: -2 = (3)^2 - 5(3) + k - 2 = 9 - 15 + k - 2 = -6 + k To solve for k$, add 6 to both sides of the equation: $-2 + 6 = k4 = kTherefore, the value of k$ is 4.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. What are the coordinates of the x-intercepts of the function?

a. **(1, 0) and (3, 0)** ✓

b. (0, 1) and (0, 3)

c. (-1, 0) and (-3, 0)

d. (0, -1) and (0, -3)

The x-intercepts of a function are the points where the graph crosses the x-axis, meaning the y-coordinate is 0. Thus, we need to solve $f(x) = 0$. \\ $x^2 - 4x + 3 = 0$ Factoring the quadratic equation, we get $(x - 1)(x - 3) = 0$ Therefore, the x-intercepts occur at $x = 1$ and $x = 3$. The coordinates of the x-intercepts are (1, 0) and (3, 0).

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. It passes through the points $(0, -3)$, $(1, 0)$, and $(-1, -4)$. Find the value of c.

a. **-3 ✓**

b. 3

c. -1

d. 1

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. The graph of the function passes through the point $(0, -3)$. Substituting these coordinates into the function, we get: $f(0) = a(0)^2 + b(0) + c$. Since $f(0) = -3$, we have $-3 = 0 + 0 + c$, which means $c = -3$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown below. Find the value of the y-intercept.

a. **5 ✓**

b. -5

c. 0

d. 1

The y-intercept of a function occurs where the graph crosses the y-axis, which is when $x = 0$. For the function $f(x) = x^2 - 6x + 5$, when $x = 0$, we have $f(0) = (0)^2 - 6(0) + 5 = 0 - 0 + 5 = 5$. Therefore, the y-intercept is 5.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A particle's position on the x-axis at time t (in seconds) is given by the function $x(t) = t^3 - 6t^2 + 9t + 1$. The particle starts at $t = 0$. Determine the time intervals during which the particle is moving in the negative direction and calculate the total distance traveled by the particle during the first 4 seconds.

a. Total distance = 10 units

b. Total distance = 12 units

c. **Total distance = 14 units ✓**

d. Total distance = 16 units

First, find the velocity function by differentiating the position function with respect to time: $v(t) = x'(t) = 3t^2 - 12t + 9$. To find when the particle is moving in the negative direction, we set $v(t) < 0$. Factor the velocity function: $v(t) = 3(t^2 - 4t + 3) = 3(t - 1)(t - 3)$. Thus, $v(t) < 0$ when $1 < t < 3$. So, the particle moves in the negative direction during the interval $(1, 3)$ seconds. Next, calculate the total distance traveled during the first 4 seconds. This requires finding the positions at criti.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider a cubic polynomial $P(x) = ax^3 + bx^2 + cx + d$. Given that $P(1) = 10$, $P(2) = 27$, $P(3) = 58$, and $P(4) = 115$, determine the value of a.

a. **1** ✓

b. 2

c. 3

d. 4

Let the polynomial be $P(x) = ax^3 + bx^2 + cx + d$. We are given the following values:

$P(1) = a(1)^3 + b(1)^2 + c(1) + d = a + b + c + d = 10$ (1) $P(2) = a(2)^3 + b(2)^2 + c(2) + d = 8a + 4b + 2c + d = 27$ (2)

$P(3) = a(3)^3 + b(3)^2 + c(3) + d = 27a + 9b + 3c + d = 58$ (3) $P(4) = a(4)^3 + b(4)^2 + c(4) + d = 64a + 16b + 4c + d = 115$ (4) Consider the differences between consecutive terms: $P(2) - P(1) = (8a + 4b + 2c + d) - (a + b + c + d) = 7a + 3b + c = 27 - 10 = 17$ (5).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Statistics & Probability

Consider a geometric progression where the sum of the first n terms is given by $S_n = \frac{3(2^n - 1)}{2}$. A second sequence, denoted by a_n, is defined such that $a_n = S_n - S_{n-1}$ for $n \geq 2$, and $a_1 = S_1$. Find the value of $a_5 - a_3$.

a. 24

b. 18

c. 48

d. **36** ✓

The sum of the first n terms of the geometric progression is $S_n = \frac{3(2^n - 1)}{2}$. The n-th term of the sequence, a_n, can be found using the relation $a_n = S_n - S_{n-1}$ for $n \geq 2$, and $a_1 = S_1$. First, let's find a_1: $a_1 = S_1 = \frac{3(2^1 - 1)}{2} = \frac{3(1)}{2} = \frac{3}{2}$. Now, let's find a general formula for a_n for $n \geq 2$: $S_{n-1} = \frac{3(2^{n-1} - 1)}{2}$. $a_n = S_n - S_{n-1} = \frac{3(2^n - 1)}{2} - \frac{3(2^{n-1} - 1)}{2}$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Calculus

The histogram below shows the number of hours 100 students spent studying for an exam. The number of students who studied for between 0 and 5 hours is represented by a bar with a height of 30. Determine the number of students who studied for between 5 and 10 hours.

a. 20 ✓

b. 25

c. 30

d. 35

The histogram displays the distribution of study hours for 100 students. The total number of students is 100. The question states that the number of students who studied for between 0 and 5 hours is 30. The bar for the '0-5 hours' interval has a height of 30, representing 30 students. The bars for the '5-10 hours' and '10-15 hours' intervals have heights of 20 and 15 respectively, representing 20 and 15 students. Therefore, the number of students who studied for between 5 and 10 hours is 20.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 AI SL — Specific Topics

Consider the equation $\log_x(81) = 4$. If x is a positive real number, find the value of x.

a. 3 ✓

b. $\sqrt{3}$

c. 9

d. $2\sqrt{3}$

The equation $\log_x(81) = 4$ can be rewritten in exponential form as $x^4 = 81$. To find x, we need to take the fourth root of 81. Since 81 can be expressed as 3^4, we have $x^4 = 3^4$. Given that x is a positive real number, the unique solution is $x = 3$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Functions

The graph of the function $f(x) = \sqrt{x-1} + 2$ is shown below. What is the value of $f(5)$?

a. 2

b. 3

c. 4 ✓

d. 5

To find the value of $f(5)$, substitute $x = 5$ into the function $f(x) = \sqrt{x-1} + 2$. $f(5) = \sqrt{5-1} + 2$ $f(5) = \sqrt{4} + 2$ $f(5) = 2 + 2$ $f(5) = 4$ Therefore, the value of $f(5)$ is 4.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Geometry & Trigonometry

How many different 4-digit codes can be formed using the digits 1-9 without repetition, if the code must be even?

a. 1344 ✓

b. 3024

c. 504

d. 672

The last digit must be 2, 4, 6, or 8 (4 choices). The remaining 3 positions are filled from the remaining 8 digits, in order: $8 \times 7 \times 6 = 336$. Total: $4 \times 336 = 1344$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Statistics & Probability

Consider the function $f(x) = 2x - 3$. Determine the value of $f(3)$.

a. 3

b. 6 ✓

c. 9

d. 0

To find the value of $f(3)$, substitute $x = 3$ into the function $f(x) = 2x - 3$. $f(3) = 2(3) - 3$ $f(3) = 6 - 3$ $f(3) = 3$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Calculus

How many ways can 4 different books be arranged on a shelf?

a. 24 ✓

b. 12

c. 16

d. 8

Applying Number & Algebra principles, $4! = 4 \times 3 \times 2 \times 1 = 24$. Therefore, the verified correct option is 24.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 AI SL — Specific Topics

Consider the function $f(x) = 2x + 5$. What is the value of $f(3)$?

a. 11 ✓

b. 8

c. 10

d. 16

To find $f(3)$, substitute $x = 3$ into the function $f(x) = 2x + 5$. $f(3) = 2(3) + 5 = 6 + 5 = 11$.

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